

Numerical Analysis of All MEPFS Model Features

Auto-generated Analysis

November 14, 2025

Distribution and Importance of All 44 Features

The following table shows the statistical distribution and model importance for all 44 features, including the 8 core inputs and the 36 degree-2 polynomial features (squared terms and pairwise interactions).

- **Min (0th Pctl):** The minimum value observed for the feature.
- **25th Pctl:** The first quartile; 25% of the data falls below this value.
- **Median (50th Pctl):** The central value of the dataset.
- **75th Pctl:** The third quartile; 75% of the data falls below this value.
- **Max (100th Pctl):** The maximum value observed for the feature.
- **Importance (%):** The feature's relative contribution to the model's predictive power.

From Raw Scores to Percentage Importance

While the raw feature importance scores (calculated by summing MSE reductions) are useful for ranking features, they are often normalized to be expressed as percentages. This makes them easier to interpret as the relative contribution of each feature to the overall model performance.

$$\text{Percentage Importance}(j) = \left(\frac{\text{Imp}(j)}{\sum_{k=1}^N \text{Imp}(k)} \right) \times 100\%$$

Feature Name	Min	25th Pctl	Median	75th Pctl	Max	Importance (%)
year	2.00×10^3	2.02×10^3	2.02×10^3	2.02×10^3	2.02×10^3	2.72
num_storeys	1.00	2.00	2.00	2.00	4.00	0.00
num_classrooms	1.00	4.00	4.00	6.00	2.00×10^1	4.10
total_electrical_cost	2.10×10^4	2.11×10^5	3.90×10^5	7.92×10^5	3.88×10^6	1.12
total_plumbing_sewer_cost	0.00	1.49×10^5	2.26×10^5	3.41×10^5	2.49×10^6	1.09
electrical_cost_per_classroom	1.71×10^3	5.74×10^4	8.54×10^4	1.52×10^5	4.80×10^5	7.30
plumbing_cost_per_classroom	0.00	3.24×10^4	4.86×10^4	7.16×10^4	2.08×10^5	3.20
fire_alarm_cost_per_classroom	0.00	0.00	6.70×10^3	1.62×10^4	2.97×10^5	3.00
year ²	4.00×10^6	4.08×10^6	4.09×10^6	4.10×10^6	4.10×10^6	0.00
year*num_storeys	2.00×10^3	4.03×10^3	4.04×10^3	4.05×10^3	8.10×10^3	2.25
year*num_classrooms	2.02×10^3	8.08×10^3	8.09×10^3	1.21×10^4	4.04×10^4	3.47
year*total_electrical_cost	4.26×10^7	4.26×10^8	7.87×10^8	1.60×10^9	7.86×10^9	0.00
year*total_plumbing_sewer_cost	0.00	3.01×10^8	4.57×10^8	6.90×10^8	5.04×10^9	0.00
year*electrical_cost_per_classroom	1.45×10^6	1.16×10^8	1.73×10^8	3.07×10^8	9.69×10^8	0.00
year*plumbing_cost_per_classroom	0.00	6.55×10^7	9.84×10^7	1.45×10^8	4.20×10^8	0.00
year*fire_alarm_cost_per_classroom	0.00	0.00	1.35×10^7	3.28×10^7	6.01×10^8	0.00
num_storeys ²	1.00	4.00	4.00	4.00	1.60×10^1	0.00
num_storeys*num_classrooms	1.00	5.00	8.00	1.60×10^1	8.00×10^1	1.13
num_storeys*total_electrical_cost	2.10×10^4	3.48×10^5	7.08×10^5	1.69×10^6	1.55×10^7	1.35
num_storeys*total_plumbing_sewer_cost	0.00	2.10×10^5	4.22×10^5	7.44×10^5	9.96×10^6	0.00
num_storeys*electrical_cost_per_classroom	1.81×10^3	9.06×10^4	1.63×10^5	3.10×10^5	1.29×10^6	1.92
num_storeys*plumbing_cost_per_classroom	0.00	5.33×10^4	9.01×10^4	1.53×10^5	8.30×10^5	3.55
num_storeys*fire_alarm_cost_per_classroom	0.00	0.00	1.34×10^4	3.47×10^4	8.91×10^5	9.30
num_classrooms ²	1.00	1.60×10^1	1.60×10^1	3.60×10^1	4.00×10^2	0.00
num_classrooms*total_electrical_cost	1.20×10^4	8.05×10^5	1.65×10^6	4.40×10^6	4.66×10^7	1.93
num_classrooms*total_plumbing_sewer_cost	0.00	4.89×10^5	1.02×10^6	2.00×10^6	2.99×10^7	1.02
num_classrooms*electrical_cost_per_classroom	2.10×10^4	1.11×10^5	3.90×10^5	7.92×10^5	3.88×10^6	0.00

Feature Name	Min	25th Pctl	Median	75th Pctl	Max	Importance (%)
num_classrooms*plumbing_cost_per_classroom	1.49×10^5	2.26×10^5	3.41×10^5	2.49×10^6	0.00	0.00
num_classrooms*fire_alarm_cost_per_classroom	0.00	3.18×10^4	7.15×10^4	1.78×10^6	2.06	2.06
total_electrical_cost^2	4.42×10^8	4.46×10^{10}	1.52×10^{11}	6.27×10^{11}	1.51×10^{13}	0.00
total_electrical_cost*total_plumbing_sewer_cost	3.61×10^{10}	8.11×10^{10}	2.89×10^{11}	9.67×10^{12}	7.00	7.00
total_electrical_cost*electrical_cost_per_classroom	1.67×10^{10}	3.07×10^{10}	9.87×10^{10}	1.26×10^{12}	6.00	6.00
total_electrical_cost*plumbing_cost_per_classroom	1.66×10^9	1.91×10^{10}	5.08×10^{10}	8.06×10^{11}	0.00	0.00
total_electrical_cost*fire_alarm_cost_per_classroom	0.00	2.58×10^9	1.52×10^{10}	3.38×10^{11}	3.50	3.50
total_plumbing_sewer_cost^2	0.00	2.21×10^{10}	5.11×10^{10}	1.16×10^{11}	6.21×10^{12}	0.00
total_plumbing_sewer_cost*electrical_cost_per_classroom	1.63×10^9	1.91×10^{10}	5.08×10^{10}	8.06×10^{11}	0.00	0.00
total_plumbing_sewer_cost*plumbing_cost_per_classroom	5.27×10^9	1.19×10^{10}	2.67×10^{10}	5.17×10^{11}	3.30	3.30
total_plumbing_sewer_cost*fire_alarm_cost_per_classroom	0.00	1.54×10^9	5.15×10^9	2.17×10^{11}	0.00	0.00
electrical_cost_per_classroom^2	2.92×10^6	3.30×10^9	7.29×10^9	2.32×10^{10}	2.30×10^{11}	0.00
electrical_cost_per_classroom*plumbing_cost_per_classroom	1.94×10^9	2.78×10^9	9.02×10^9	6.72×10^{10}	8.40	8.40
electrical_cost_per_classroom*fire_alarm_cost_per_classroom	0.00	3.85×10^8	2.53×10^9	3.33×10^{10}	0.00	0.00
plumbing_cost_per_classroom^2	0.00	1.05×10^9	2.37×10^9	5.12×10^9	4.31×10^{10}	0.00
plumbing_cost_per_classroom*fire_alarm_cost_per_classroom	0.00	2.68×10^8	9.74×10^8	2.17×10^{10}	0.00	0.00
fire_alarm_cost_per_classroom^2	0.00	0.00	4.49×10^7	2.64×10^8	8.82×10^{10}	0.00